SMART AGRICULTURE AGENT

AI-Based Crop Disease Detection & Irrigation Advisory System

B.Tech Computer Science & Engineering Mini-Capstone Project • Four-Member Team

Problem Statement: Foliar diseases and unscientific irrigation reduce global crop yields by 20-40%. Smallholder farmers lack rapid laboratory diagnostics, leading to pesticide over-application and groundwater water waste.

Project Objective: Deliver an edge-capable agricultural decision-support system coupling deep transfer learning (MobileNetV2) for foliar classification with deterministic FAO MAD irrigation water balance rules.

AI Technique (MobileNetV2):
• **Inverted Residuals:** Depthwise separable convolutions maximize feature representation on CPU.
• **Transfer Learning:** ImageNet backbone fine-tuned on PlantVillage (8 classes across Tomato, Potato, Corn).
• **Head:** Dense(1280→128) → ReLU → Dropout(0.3) → Dense(8).
• **Confidence Gating:** Threshold (τ = 70%) withholds unverified chemical prescriptions.

Irrigation Rule Engine (FAO-56):
• **MAD Depletion:** Tomato (35%), Potato (40%), Corn (30%).
• **Soil Capacity:** Sandy (+6% trigger), Clayey (-5% trigger).
• **Precipitation Override:** Rain ≥ 12 mm in 24h defers watering to prevent root rot and hypoxia.
• **Data Integrity:** Missing sensor values are never assumed zero.

ACTUAL MEASURED EXPERIMENTAL RESULTS (Held-Out Test Set)

Test Accuracy	Macro F1-Score	Mean CPU Latency	Automated Pytest
85.19%	**0.8333**	**25.75 ms**	**31/31 Passed**

Implementation & Stack: Python 3.12, PyTorch CPU 2.14, Flask 3.1, SQLite, Pillow, Scikit-learn, HTML5/CSS3. Modular blueprints, parameterized SQL queries, and zero hardcoded secrets.

Four-Member Team Distribution:
• **Member 1 (AI/Vision):** Dataset curation, MobileNetV2 fine-tuning, test evaluation.
• **Member 2 (Decision Engines):** Knowledge base, gating rules, FAO irrigation engine.
• **Member 3 (Full-Stack):** Flask architecture, responsive UI, database persistence.
• **Member 4 (QA & Mentorship):** Pytest suite (31 tests), latency audits, poster/report.

Responsible AI & Limitations:
• **Safe Decision Support:** Model confidence is explicitly labeled; synthetic chemical spray dosages are strictly withheld.
• **Crop Mismatch Warning:** Cross-checks farmer crop vs leaf feature prediction.
• **Limitations:** Laboratory images lack field background clutter; soil pathogens require in-person extension ground verification.

Conclusion: Successfully demonstrates an explainable, CPU-efficient AI assistance agent with verified 85.2% accuracy and sub-30ms latency.

References: PlantVillage (Hughes & Salathé, 2015) • MobileNetV2 (Sandler et al., 2018) • FAO-56 Irrigation • UC IPM Guidelines.